

Exercise series 3: Solutions

Introduction to databases



ADReM
Advanced Database Research & Modelling
University of Antwerp

 **Universiteit
Antwerpen**

Exercise 1

RelaX > Load dataset > **Database Systems The Complete Book - Exercise 2.4.1**

1. Find the maker, model and price of all PC's costing less than 1000 euro.
2. Find all makers that make PC's.
3. Find all makers that sell PC's or Laptop's with at least 2 Gb of ram.
4. Find all makers that do **not** make laptops or printers.
5. ! Find all **pairs** of makers that produce a type in common.
(A pair should be listed only once; e.g., list (i, j) but not (j, i).)
6. Find the model number and price of all products (of any type) made by manufacturer B.
7. Find those manufacturers that sell Laptops, but not PC's.
8. ! Find those hard-disk sizes that occur in two or more PC's.
9. ! Find those pairs of PC models that have both the same speed and RAM.
10. !! Find those manufacturers of at least two different computers (PC's or laptops) with speed of at least 2.80.
11. !! Find the manufacturer(s) of the computer (PC or laptop) with the highest available speed.
12. !! Find the manufacturers of PC's with at least three different speeds.
13. !! Find the manufacturers who sell exactly three different models of PC.



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Exercise 1

1. Find the maker, model and price of all PC's costing less than 1000 euro.
 $\pi \text{ maker, model, price (Product } \bowtie \sigma \text{ price} < 1000 \text{ PC)}$
2. Find all makers that make PC's.
 $\pi \text{ maker } \sigma \text{ type} = \text{'pc'} \text{ Product}$
3. Find all makers that sell PC's or Laptop's with at least 2 Gb of ram.
 $\pi \text{ maker (Product } \bowtie \sigma \text{ ram} > 2000 (\pi \text{ model, ram PC } \cup \pi \text{ model, ram Laptop}))$
4. Find all makers that do **not** make laptops or printers.
 $\pi \text{ maker Product} - \pi \text{ maker } (\sigma \text{ type} = \text{'laptop'} \text{ Product } \cup \sigma \text{ type} = \text{'printer'} \text{ Product})$
5. ! Find all **pairs** of makers that produce a type in common.
 $\sigma \text{ maker2} < \text{maker } (\rho \text{ maker2 } \leftarrow \text{maker } \pi \text{ maker, type Product } \bowtie \pi \text{ maker, type Product})$
6. Find the model number and price of all products (of any type) made by manufacturer *B*.
 $\pi \text{ model, price } ((\pi \text{ model, price Laptop } \cup \pi \text{ model, price PC } \cup \pi \text{ model, price Printer}) \bowtie \sigma \text{ maker} = \text{'B'} \text{ Product})$



Exercise 1

7. Find those manufacturers that sell Laptops, but not PC's.
 $\pi \text{ maker } \sigma \text{ type} = \text{'laptop'} \text{ Product} - \pi \text{ maker } \sigma \text{ type} = \text{'pc'} \text{ Product}$
8. ! Find those hard-disk sizes that occur in two or more PC's.
 $\pi \text{ hd } \sigma \text{ model} < \text{model2 } (\pi \text{ model, hd PC } \bowtie \rho \text{ model2 } \leftarrow \text{model } \pi \text{ model, hd PC})$
9. ! Find those pairs of PC models that have both the same speed and RAM.
 $\pi \text{ model, model2 } \sigma \text{ model} < \text{model2 } (\pi \text{ model, speed, ram PC } \bowtie \rho \text{ model2 } \leftarrow \text{model } \pi \text{ model, speed, ram PC})$
10. !! Find those manufacturers of at least two different computers (PC's or laptops) with speeds of at least 2.80.
 **$A = (\pi \text{ model, maker Product } \bowtie \pi \text{ model } \sigma \text{ speed } \geq 2.8 (\pi \text{ model, speed PC } \cup \pi \text{ model, speed Laptop}))$
 $\pi \text{ maker } \sigma \text{ model2 } \neq \text{model } (\rho \text{ model2 } \leftarrow \text{model } A \bowtie A)$**



Exercise 1

11. !! Find the manufacturer(s) of the computer (PC or laptop) with the highest available speed.

$A = (\pi \text{ model, speed PC} \cup \pi \text{ model, speed Laptop})$

$\pi \text{ maker (Product} \bowtie (\pi \text{ model } A - \pi \text{ model } \sigma \text{ speed} < \text{speed2 (} A \times \rho \text{ model2} \leftarrow \text{model, speed2} \leftarrow \text{speed } A)))$

12. !! Find the manufacturers of PC's with at least three different speeds.

$A = (\pi \text{ maker, speed (PC} \bowtie \text{Product}))$

$\pi \text{ maker } \sigma \text{ speed2} < \text{speed3 (} \rho \text{ speed3} \leftarrow \text{speed } A \bowtie \sigma \text{ speed} < \text{speed2 (} A \bowtie \rho \text{ speed2} \leftarrow \text{speed } A))$

13. !! Find the manufacturers who sell exactly three different models of PC.

$A = (\pi \text{ maker, model (Product} \bowtie \text{PC}))$

$\pi \text{ maker } \sigma \text{ model1} < \text{model2} \wedge \text{model2} < \text{model3}$

$(\rho \text{ model1} \leftarrow \text{model } A \bowtie \rho \text{ model2} \leftarrow \text{model } A \bowtie \rho \text{ model3} \leftarrow \text{model } A)$

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$\pi \text{ maker } \sigma \text{ model1} < \text{model2} \wedge \text{model2} < \text{model3} \wedge \text{model3} < \text{model4}$

$(\rho \text{ model1} \leftarrow \text{model } A \bowtie \rho \text{ model2} \leftarrow \text{model } A \bowtie \rho \text{ model3} \leftarrow \text{model } A \bowtie \rho \text{ model4} \leftarrow \text{model } A)$



Exercise 2: Social Media

User

User.email	User.firstname	User.lastname	User.age
len.feremans@uantwerpen.be	len	feremans	37
toon.calders@uantwerpen.be	toon	calders	44
koen.wouters@gmail.be	koen	wouter	52
peter.selle@uantwerpen.be	Peter	Selle	20
spin.azie@uantwerpen.be	Spin	Azie	20
potus@gov.us	Donald	Trump	65

Friends

Friends.user1	Friends.user2	Friends.friends_o
toon.calders@uantwerpen.be	len.feremans@uantwerpen.be	2015-01-01
toon.calders@uantwerpen.be	koen.wouters@gmail.be	2016-07-13
len.feremans@uantwerpen.be	peter.selle@uantwerpen.be	2015-02-10
toon.calders@uantwerpen.be	peter.selle@uantwerpen.be	2015-09-03

Message

Message.creator	Message.created_on	Message.body
len.feremans@uantwerpen.be	2018-10-11	hello world!
len.feremans@uantwerpen.be	2018-10-12	Creating a new Social Network schema!
peter.selle@uantwerpen.be	2018-10-11	--Database4Ever!

RelaX > Load dataset

> Load dataset stored in a gist > Copy-paste

4d7c194588837bc8cd3be1491a4d2450

MyGroup

MyGroup.name	MyGroup.admin
UAntwerpen	toon.calders@uantwerpen.be
Republicans	potus@gov.us

Member

Member.user_email	Member.group_name
toon.calders@uantwerpen.be	UAntwerpen
len.feremans@uantwerpen.be	UAntwerpen
peter.selle@uantwerpen.be	UAntwerpen

Likes

Likes.liker_user_email	Likes.m_creator
peter.selle@uantwerpen.be	len.feremans@uantwerpen.be
toon.calders@uantwerpen.be	len.feremans@uantwerpen.be
toon.calders@uantwerpen.be	len.feremans@uantwerpen.be
len.feremans@uantwerpen.be	peter.selle@uantwerpen.be

Exercise 2: Social media

1. Which users created a message that is liked ?
2. Which users that are 20, have friends that are also 20?
3. Which friends do Toon and Len share?
4. Who likes messages created by members of the group UAntwerpen?
5. Which friend of Toon does not like any message?
6. Which friend of Toon likes all messages?



Exercise 2: Social media

1. Which users created a message that is liked ?

$\pi m_creator (Likes)$

2. Which users that are 20, have friends that are also 20?

$A = (\pi email \sigma age = 20 User)$

$SymF = (Friends \cup \pi user2, user1, friends_on Friends)$

$\pi user1 (\rho user1 \leftarrow email A \bowtie SymF \bowtie \rho user2 \leftarrow email A)$

3. Which friends do Toon and Len share?

$SymF = (Friends \cup \pi user2, user1, friends_on Friends)$

$\pi user2 \sigma user1 = 'toon.calders@uantwerpen.be' SymF \cap \pi user2 \sigma user1 = 'len.feremans@uantwerpen.be' SymF$

4. Who likes messages created by members of the group UAntwerpen?

$Uantwerpen_Members = \pi user_email \sigma group_name = 'UAntwerpen' Member$

$\pi liker_user_email (\rho m_creator \leftarrow user_email Uantwerpen_Members \bowtie Likes)$

5. Which friend of Toon does not like any message?

$toon_friends = (\pi user2 \sigma user1 = 'toon.calders@uantwerpen.be' Friends \cup \pi user1 \sigma user2 = 'toon.calders@uantwerpen.be' Friends)$

$toon_friends - \pi liker_user_email Likes$

6. Which friend of Toon likes all messages?

$(\rho creator \leftarrow m_creator, created_on \leftarrow m_created_on Likes \div \pi creator, created_on Message) \cap toon_friends$

